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DEALER BEST PRACTICE SERIES

Salvage Mounting Surfaces of the Track Guides

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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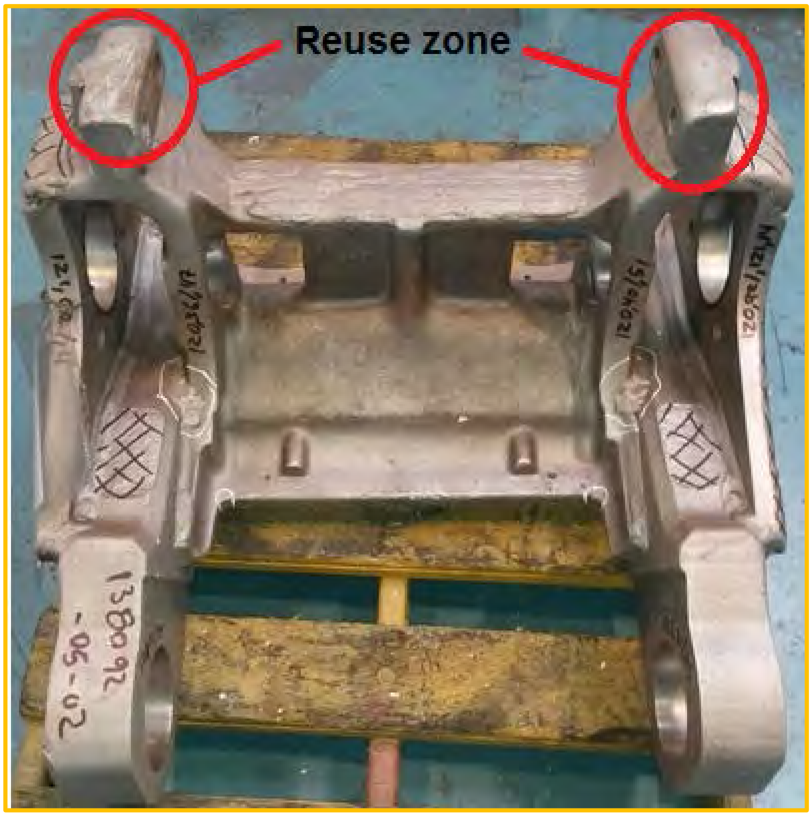
1.0 Introduction

According to Reuse and Salvage guidelines REHS1757, worn or damaged skids from Bogie Front, Bogie Middle and Bogie Rear for models D10N, D10R, D10T, D10T2, and D11N, D11R, D11T can be salvaged by welding.

Ferreyros has identified a process that can reduce time when there is excessive worn or damaged skids.

2.0 Best Practice Description



Picture 1: Bogie Middle, skids showing worn.

Picture 2: Bogie Front, skid presents excessive damage

Previously, the recovery of excessive worn skids is as follows:

- Remove skids with excessive damage.
- Manufacture two rectangular blocks with dimensions similar to the removed skid
- Weld blocks.
- Grinding blocks according to print.

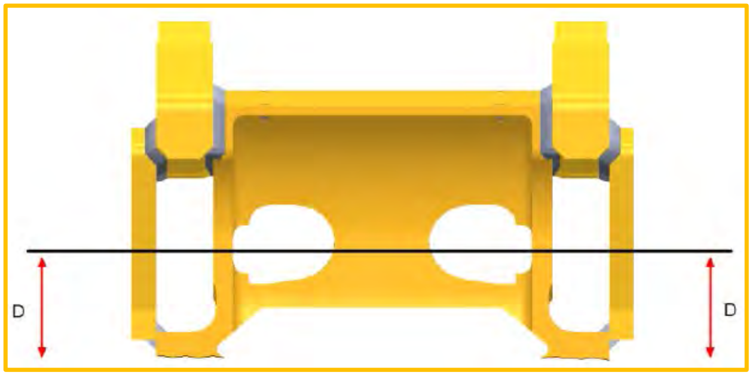
This process takes approximately 76.43 hours and the component stays in the shop for about 20 days.

Currently, skids replacement are made, which will be centered by two dowels. These skids are to finished size and welding is only needed for grinding the surface wears.

With this improvement, time to perform this process was reduced from 76.43 to 58.434, while the time the component stays in the workshop decreases by 10 days.

3.0 Implementation Steps

- Evaluate Bogies as measures indicated on the print to see the need to change skids.
- Remove worn skates with Arc- Air. To remove skates considering Table 1 and Figure 3.



Picture 3: Distance to cut excessive wear skate

Table 1

Bogie	Length D (mm)
D10	140 mm
D11	160 mm

- Rectify the area that was cut with Arc-Air; see final dimensions in Table 2.

Note: For grinding, place the Bogie in a device (Picture 4). In case Bogie presents threaded damaged bores, they have to be corrected because the component has to be centered in the device by this threaded bores.

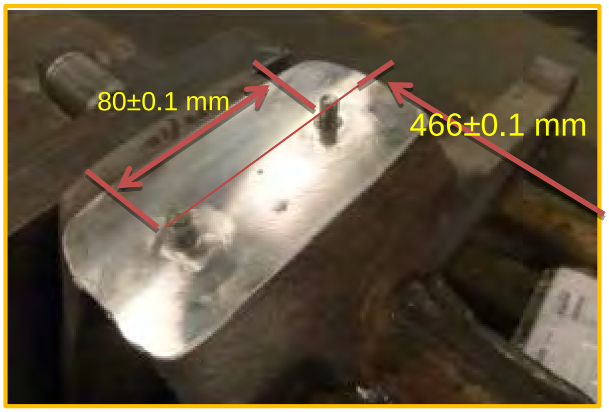
Tabla 2

Bogie	Length D (mm)
D10	125 mm
D11	148 mm

Picture 4: Device where Bogies are housed

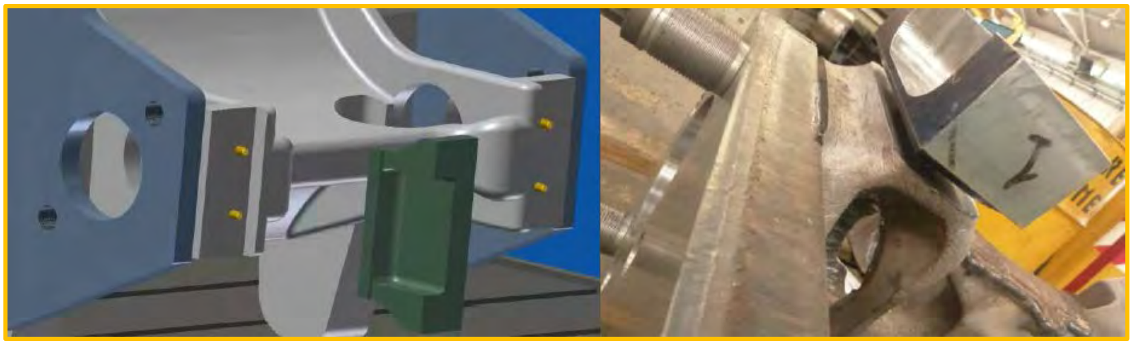
- Drill holes for guide pins and install the pins. These pins will center new skids.

  NOTE: Drill a depth of 10 mm. hole spacing is 80±0.1 mm while the hole spacing between the skids is 466 ± 0.1 mm.



Picture 5: Distance between holes for pins on skids

- Install new skids taking as reference the location of the guide pin.



Picture 6: Installation of new skids

- Weld the new skids mounted on the Bogie:

  a) Preheat to 160°C the weld area (3 "around).

  b) Use welding process FCAW and tubular wire AWS A5.20. Strutting installed skids, then weld opposite zones to avoid deformation.

     NOTE: Pay attention to manufacturing best practices for the welding process.

  c) Cover with blankets, finish and cool.

- Standardize welded area using a grinding wheel.

4.0 Benefits

- Reduced turnaround time by 45%.
- Reduced hours in 23.6%.

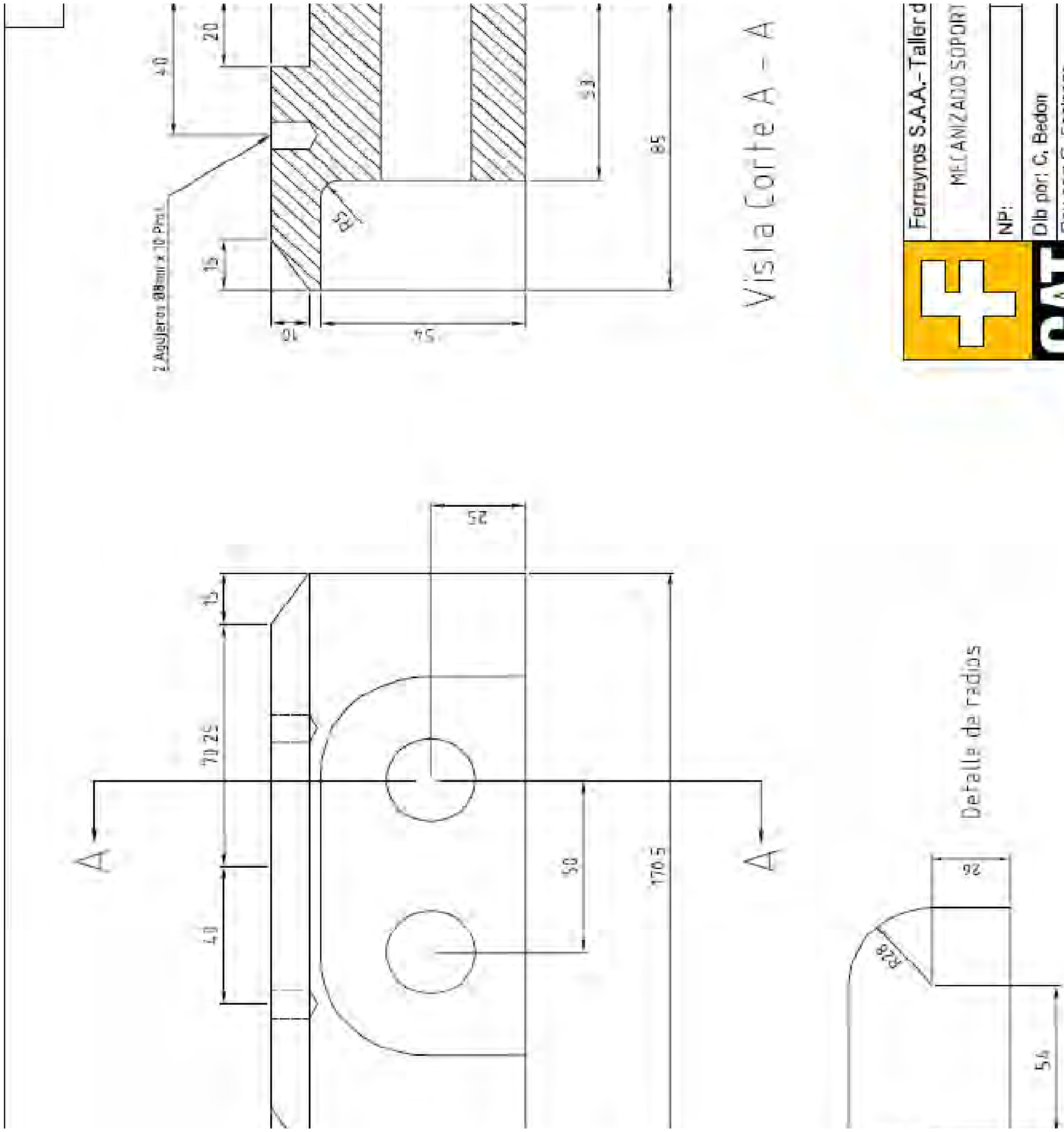
  i. Preceding process, 76.433 h
  ii. Current process: 58.434 h

5.0 Resources Required

- Enabled skids.
- Standard AWS D1.1 2010

6.0 Supporting Attachments / References

A. Enabled skids D10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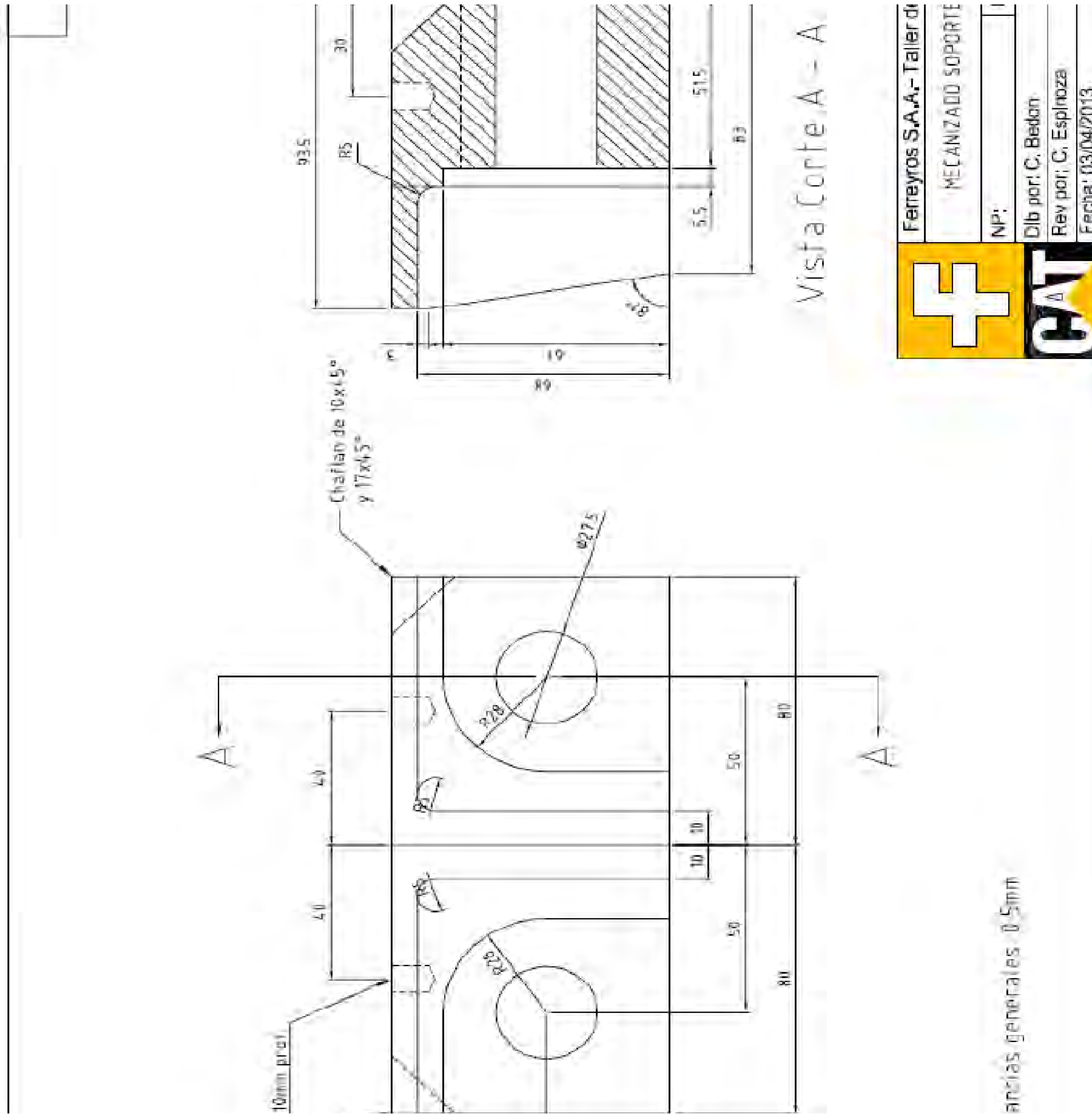
B. Enabled skids D11



7.0 Related Best Practices

None

8.0 Acknowledgements

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